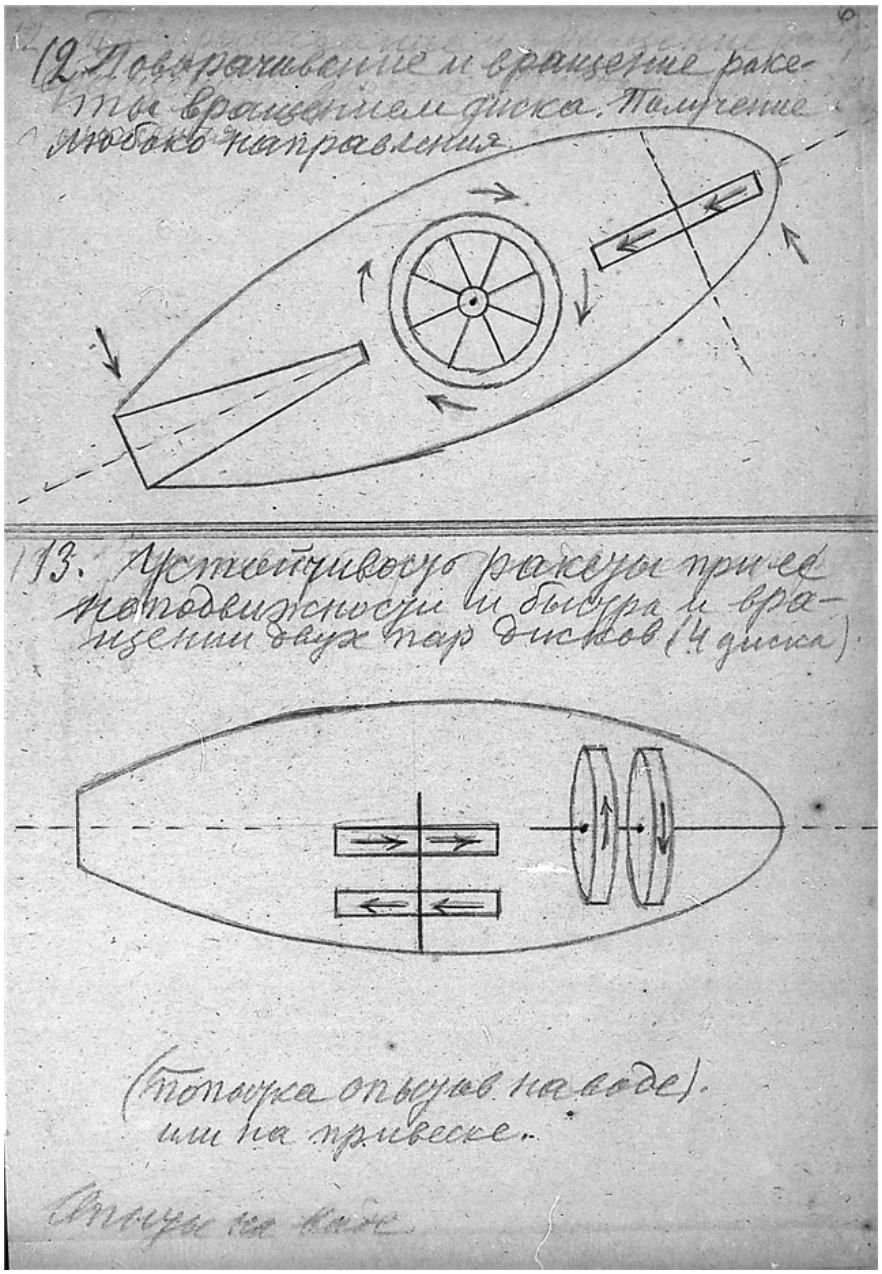
0. Write down the names of any people you worked with and any resources (online or otherwise) that you used.

Reaction wheels

Spacecraft, such as telescopes, need to be able to rotate in 3D to point at their various targets. They can rotate using thrusters, but this uses fuel, which is non-renewable. Spacecraft don't tend to get refueled, so once they're out of fuel they're dead. Alternatively, they can rotate using reaction wheels. Reaction wheels use electricity, which can be replenished for free using solar panels. What are reaction wheels? Take a look at the top half of the figure below, then read on.



As illustrated in the top figure, the spacecraft has a large reaction wheel in the middle. By spinning the reaction wheel clockwise, the spacecraft itself will spin counterclockwise, as illustrated

by the arrows outside the spacecraft. Looking carefully, you will notice a second reaction wheel on the right end of the spacecraft. It's perpendicular to the one in the middle. According to the arrows drawn on it, is turning to the left from our perspective. Its axis is the dashed line drawn going up and down at a slight angle. Why the second wheel? If you only have one reaction wheel, you can only do 2D rotations about the axis of that wheel. If you want 3D rotations, you need more wheels.

The figure above was published by Tsiolkovsky, an early rocket scientist, in 1933, which is notable because spacecraft did not exist in 1933. As for the lower figure, it shows how you can use pairs of wheels spinning in opposite directions to stabilize the spacecraft against outside forces that try to rotate it, which is also used today more or less.

1. Focusing on the top figure above, we will define axes as follows. The axis of the wheel in the middle is the y-axis, going in and out of the page. The axis of the wheel on the right is the z-axis, going almost up and down (at a slight angle).

   What about the x-axis? It seems like there's a missing reaction wheel. Add it to the drawing.

   Solution: On the drawing, the third wheel should look like the one on the right, but it should be perpendicular. It doesn't matter where on the spacecraft you draw it.

2. Maybe Tsiolkovsky was right all along, and the spacecraft can ditch the third wheel? What is the result of rotating by 180° using the middle reaction wheel (y-axis), followed by rotating by 180° using the other reaction wheel (z-axis)?

   Grab a handy object and try it out so that you know what answer you should get, and then use what you've learned about 3D rotations in the previous homework to compose the two rotations using quaternions.

   This problem is forgiving if you're not careful about which order you multiply quaternions, but you should try to get it right anyways to be ready for later problems.

   Solution: We saw earlier that a 180° rotation about the x-axis is represented by the quaternions $\pm\mathbf{i}$. Similarly, a 180° rotation about the y-axis is represented by the quaternions $\pm\mathbf{j}$, and a 180° rotation about the z-axis is represented by the quaternions $\pm\mathbf{k}$.

   Thus, we can represent our first rotation by $\mathbf{j}$, and our second rotation by $\mathbf{k}$. As we saw in the section on composing 3D rotations in the previous homework, the result is given by the product of the two quaternions, with the *second* quaternion written first. Thus, the resulting rotation is given by the quaternion

   $$\mathbf{kj} = -\mathbf{i},$$

   which is one of the two quaternions representing a 180° rotation about the x-axis. Thus, we can do a 180° rotation about the x-axis without an x-axis reaction wheel; the two wheels in the original drawing are enough.

Some more 3D examples

3. Write down all quaternions that yield a 90° rotation about a coordinate axis.

Keep in mind that rotations can be clockwise or counterclockwise; include both. Also keep in mind that each rotation is represented by two quaternions; include both. Arrange your answers in a sensible way, making it clear in some way which pairs of quaternions represent the same rotation, which rotations are about the x-axis, which are about the y-axis, and which are about the z-axis. Don't forget about the 2 in 2θ.

Solution: Recall that $r = e^{\mathbf{u}\,\theta} = \cos\theta + \mathbf{u}\sin\theta$ represents a rotation by 2θ about the axis $\mathbf{u}$. In this case, $2\theta = 90°$, so $\theta = 45°$, and so we have

$$r = \tfrac{\sqrt{2}}{2} + \mathbf{u}\,\tfrac{\sqrt{2}}{2}.$$

To deal with the clockwise and counterclockwise possibilities, one option is to allow $\mathbf{u}$ to be $\mathbf{i}$, $\mathbf{j}$, or $\mathbf{k}$, and also consider $2\theta = -90°$. An equivalent option is to only consider $2\theta = 90°$ as above, but allow $\mathbf{u}$ to be $\pm\mathbf{i}$, $\pm\mathbf{j}$, or $\pm\mathbf{k}$. This corresponds to the fact that a rotation that appears clockwise looking from above will appear counterclockwise looking from below. Either way, we get the following table of quaternions representing 90° rotations about an axis.

x-axis rotations	y-axis rotations	z-axis rotations
$\frac{\sqrt{2}}{2} + \mathbf{i}\,\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2} + \mathbf{j}\,\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2} + \mathbf{k}\,\frac{\sqrt{2}}{2}$
$\frac{\sqrt{2}}{2} - \mathbf{i}\,\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2} - \mathbf{j}\,\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2} - \mathbf{k}\,\frac{\sqrt{2}}{2}$

As we saw earlier, r and $-r$ represent the same rotation, so the above 90° rotations can also be represented by the quaternions below, with corresponding cells of the table corresponding to the same rotation.

x-axis rotations	y-axis rotations	z-axis rotations
$-\frac{\sqrt{2}}{2} - \mathbf{i}\,\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2} - \mathbf{j}\,\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2} - \mathbf{k}\,\frac{\sqrt{2}}{2}$
$-\frac{\sqrt{2}}{2} + \mathbf{i}\,\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2} + \mathbf{j}\,\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2} + \mathbf{k}\,\frac{\sqrt{2}}{2}$

Note that from a particular point of view, these are actually $-270°$ rotations. But, from another point of view, $-270°$ and 90° are the same angle.

More on spacecraft rotation

One thing to keep in mind with spacecraft rotation is the distinction between the *lab frame* and the *body frame*. The easiest way to understand the distinction is to listen to the following Beatles song: `https://www.youtube.com/watch?v=wsRatIMUSu8`.

> But the fool on the hill
> Sees the sun going down
> And the eyes in his head
> See the world spinning round

According to the *lab frame*, the Earth is spinning; when we watch a sunset, what is happening is that we are moving away from the sun, eventually spinning around behind the Earth to where we can no longer see it. Meanwhile, according to the *body frame*, the sky is spinning, which causes the sun in the sky to move below the horizon and set; the spinning of the sky also causes the stars to move, as illustrated in the following long exposure from Wikipedia.

Figure 1: `https://commons.wikimedia.org/wiki/File:All_In_A_Spin_Star_trail.jpg`

While it is unequivocally true that the Earth is spinning, not the sky, in most circumstances it is actually the *body frame* that is more intuitive and more useful. After all, we usually think about a sunset as the sun going down, not as us moving away from it.

In terms of spacecraft, the body frame means we take the perspective of someone on the spacecraft looking out the window and seeing everything outside spin around them; from this perspective, the spacecraft does not move. If we are a telescope and want to look at something, from the body frame perspective, we *spin the sky* until the object we want to look at is in our field of view.

The reason this perspective is usually more useful is because if the spacecraft's main engines start out pointing in the negative x direction, from the perspective of the body frame they will always point in the negative x direction, which makes it easy to compute what happens if we fire the engines. From this perspective, the universe is moving, not the spacecraft.

The same goes for reaction wheels. If we have reaction wheels around the y and z axes, then, using the body frame, they stay on the y and z axes, which makes it much much easier to compute what happens if you spin the reaction wheels; the universe moves, but the spacecraft and the reaction wheels do not. If we used the lab frame, we would also have to keep track of how the axis of one wheel changes as we spin the other wheel, which would be much more complicated.

So, the main upshot of this discussion is that you don't have to worry about the axes of the reaction wheels changing, as long as you take the perspective of the body frame: the stars in the sky are rotating, not the spacecraft.

Computing spacecraft rotations

Using the body frame takes some care. Referring back to Tsiolkovsky's figure, if we spin the center reaction wheel as shown, the wheel spins clockwise from our perspective, which means that, in the lab frame, the spacecraft spins in the opposite direction, counterclockwise, as shown. However, in

the body frame, the spacecraft does not spin; instead, the universe spins *clockwise*.

In summary, spinning the middle reaction wheel in the direction shown until the sky spins 90° will send a star on the positive z-axis to the positive x-axis, and it will send a star on the positive x-axis to the negative z-axis. Clockwise rotations are represented using negative angles, but note also that it is the negative y-axis that is pointing towards us. Thus, to represent this rotation using quaternions, for a 90° clockwise rotation about the negative y-axis, we should use $2\theta = -90°$ and $\mathbf{u} = -\mathbf{j}$. Equivalently, we can view it as a counterclockwise rotation about the positive y-axis, which is represented using $2\theta = 90°$ and $\mathbf{u} = \mathbf{j}$.

4. Verify the above paragraph. Use Question 3 as a helpful reference. Detailed instructions of what you need to do are below.

   First, write down a quaternion r representing the rotation using the values of 2θ and $\mathbf{u}$ above. (You should get the same r regardless of which of the two options for 2θ and $\mathbf{u}$ you use.)

   Solution: Based on the table in Question 3 and being careful about signs, we find

   $$r = \tfrac{\sqrt{2}}{2} + \mathbf{j}\,\tfrac{\sqrt{2}}{2}.$$

   Then, use the quaternion rotation formula $r\,\mathbf{v}\,\overline{r}$ to do the following.

   (a) Show that if $\mathbf{v}$ represents a star on the positive z-axis, then the result after rotation will be on the positive x-axis.

   Solution: Let $\mathbf{v} = a\mathbf{k}$, where $a > 0$. (If you just took $\mathbf{v} = \mathbf{k}$, that's fine.) Then

   $$\begin{aligned} r\,\mathbf{v}\,\overline{r} &= \left(\tfrac{\sqrt{2}}{2} + \mathbf{j}\,\tfrac{\sqrt{2}}{2}\right)(a\,\mathbf{k})\left(\tfrac{\sqrt{2}}{2} - \mathbf{j}\,\tfrac{\sqrt{2}}{2}\right) \\ &= a\left(\mathbf{k}\tfrac{\sqrt{2}}{2} + \mathbf{j}\,\mathbf{k}\tfrac{\sqrt{2}}{2}\right)\left(\tfrac{\sqrt{2}}{2} - \mathbf{j}\,\tfrac{\sqrt{2}}{2}\right) \\ &= a\left(\mathbf{k}\tfrac{\sqrt{2}}{2} + \mathbf{i}\tfrac{\sqrt{2}}{2}\right)\left(\tfrac{\sqrt{2}}{2} - \mathbf{j}\,\tfrac{\sqrt{2}}{2}\right) \\ &= a\left(\mathbf{k}\,\tfrac{1}{2} - \mathbf{k}\,\mathbf{j}\,\tfrac{1}{2} + \mathbf{i}\,\tfrac{1}{2} - \mathbf{i}\,\mathbf{j}\,\tfrac{1}{2}\right) \\ &= \tfrac{1}{2}a\,(\mathbf{k} + \mathbf{i} + \mathbf{i} - \mathbf{k}) \\ &= a\,\mathbf{i}, \end{aligned}$$

   which is indeed on the positive x-axis.

   (b) Show that if $\mathbf{v}$ represents a star on the positive x-axis, then the result after rotation will be on the negative z-axis.

   Solution: Now let $\mathbf{v} = a\,\mathbf{i}$. This time we will factor out the $\frac{\sqrt{2}}{2}$ factors to make things

easier.

$$\begin{aligned} r\,\mathbf{v}\,\overline{r} &= \left(\tfrac{\sqrt{2}}{2} + \mathbf{j}\,\tfrac{\sqrt{2}}{2}\right)(a\,\mathbf{i})\left(\tfrac{\sqrt{2}}{2} - \mathbf{j}\,\tfrac{\sqrt{2}}{2}\right) \\ &= \left(\tfrac{\sqrt{2}}{2}\right) a \left(\tfrac{\sqrt{2}}{2}\right)(1+\mathbf{j})(\mathbf{i})(1-\mathbf{j}) \\ &= \tfrac{a}{2}(\mathbf{i}+\mathbf{j}\,\mathbf{i})(1-\mathbf{j}) \\ &= \tfrac{a}{2}(\mathbf{i}-\mathbf{k})(1-\mathbf{j}) \\ &= \tfrac{a}{2}(\mathbf{i}-\mathbf{i}\,\mathbf{j}-\mathbf{k}+\mathbf{k}\,\mathbf{j}) \\ &= \tfrac{a}{2}(\mathbf{i}-\mathbf{k}-\mathbf{k}-\mathbf{i}) \\ &= -a\,\mathbf{k}, \end{aligned}$$

which is indeed on the negative z-axis.

Now let's consider the right reaction wheel, spinning as shown. If we look at the wheel from above (the positive z-axis), we will see it spinning clockwise. Once again, in the lab frame, the spacecraft will spin the opposite direction counterclockwise. But, in the body frame, the spacecraft will not spin, and instead the stars will spin clockwise. Thus, after spinning the reaction wheel in the direction shown until the sky spins 90°, what we will see is that a star on the positive y-axis will end up on the positive x-axis, and a star on the positive x-axis will end up on the negative y-axis. In terms of quaternions, we should represent this clockwise rotation about the positive z-axis using $2\theta = -90°$ and $\mathbf{u} = \mathbf{k}$.

5. Verify the above paragraph as before. First, write down a quaternion representing this rotation.

Solution: Being careful with signs, the quaternion is

$$r = \tfrac{\sqrt{2}}{2} - \mathbf{k}\,\tfrac{\sqrt{2}}{2}.$$

Then, use the formula $r\,\mathbf{v}\,\overline{r}$ to do the following.

(a) Show that if $\mathbf{v}$ represents a star on the positive y-axis, then the result after rotation will be on the positive x-axis.

Solution: Let $\mathbf{v} = a\,\mathbf{j}$. Then we compute as above, skipping some steps now that we get the idea.

$$\begin{aligned} r\,\mathbf{v}\,\overline{r} &= \left(\tfrac{\sqrt{2}}{2} - \mathbf{k}\,\tfrac{\sqrt{2}}{2}\right)(a\,\mathbf{j})\left(\tfrac{\sqrt{2}}{2} + \mathbf{k}\,\tfrac{\sqrt{2}}{2}\right) \\ &= \tfrac{a}{2}(1-\mathbf{k})(\mathbf{j})(1+\mathbf{k}) \\ &= \tfrac{a}{2}(\mathbf{j}+\mathbf{i})(1+\mathbf{k}) \\ &= \tfrac{a}{2}(\mathbf{j}+\mathbf{i}+\mathbf{i}-\mathbf{j}) \\ &= a\,\mathbf{i}, \end{aligned}$$

which is on the positive x-axis.

(b) Show that if $\mathbf{v}$ represents a star on the positive x-axis, then the result after rotation will be on the negative y-axis.

Solution: As before, let $\mathbf{v} = a\,\mathbf{i}$, and we compute

$$\begin{aligned} r\,\mathbf{v}\,\overline{r} &= \left(\tfrac{\sqrt{2}}{2} - \mathbf{k}\,\tfrac{\sqrt{2}}{2}\right)(a\,\mathbf{i})\left(\tfrac{\sqrt{2}}{2} + \mathbf{k}\,\tfrac{\sqrt{2}}{2}\right) \\ &= \tfrac{a}{2}(1-\mathbf{k})(\mathbf{i})(1+\mathbf{k}) \\ &= \tfrac{a}{2}(\mathbf{i}-\mathbf{j})(1+\mathbf{k}) \\ &= \tfrac{a}{2}(\mathbf{i}-\mathbf{j}-\mathbf{j}-\mathbf{i}) \\ &= -a\,\mathbf{j}. \end{aligned}$$

which is on the negative y-axis.

Earlier, we considered rotating by 180° using the middle wheel, followed by rotating 180° using the right wheel. Doing these two rotations one after the other ended up being equivalent to doing a single 180° rotation about the x-axis. We're now going to do the same thing with 90° rotations, so you can reuse your work from questions 4 and 5. In the following questions, let Rotation 1 be the rotation in question 4, a 90° rotation using the middle wheel. Let Rotation 2 be the rotation in question 5 representing a 90° rotation using the right wheel. If we apply Rotation 1 followed by Rotation 2, the result can be expressed as a single rotation. What is the axis and angle of this rotation?

We will answer this question in two different ways. In this assignment, we'll do it by hand without quaternions. Next week, we'll do it with quaternions. Remember that you can always take an object like a book and rotate it to see what's going on.

6. (a) Complete the following chart, where Rotation 3 is Rotation 1 followed by Rotation 2.

	positive x-axis	$\mapsto$	negative z-axis
Rotation 1:	positive y-axis	$\mapsto$	positive y-axis
	positive z-axis	$\mapsto$	positive x-axis
	positive x-axis	$\mapsto$	negative y-axis
Rotation 2:	positive y-axis	$\mapsto$	positive x-axis
	positive z-axis	$\mapsto$	positive z-axis
	positive x-axis	$\mapsto$	
Rotation 3:	positive y-axis	$\mapsto$	
	positive z-axis	$\mapsto$	

Hint: To apply Rotation 3 to the positive x-axis, we first apply Rotation 1, ending up at the negative z-axis. Then we apply Rotation 2 to the negative z-axis. What do we end up with?

Solution: Applying Rotation 1 to the positive x-axis, we end up with the negative z-axis. Applying Rotation 2 to the negative z-axis, we obtain the negative z-axis.

Applying Rotation 1 to the positive y-axis, we obtain the positive y-axis. Applying Rotation 2 to the positive y-axis, we obtain the positive x-axis.

Applying Rotation 1 to the positive z-axis, we obtain the positive x-axis. Applying Rotation 2 to the positive x-axis, we obtain the negative y-axis.

We write these results in the chart.

$$\text{Rotation 3:}\quad \begin{array}{ccc} \text{positive } x\text{-axis} & \mapsto & \text{negative } z\text{-axis} \\ \text{positive } y\text{-axis} & \mapsto & \text{positive } x\text{-axis} \\ \text{positive } z\text{-axis} & \mapsto & \text{negative } y\text{-axis} \end{array}$$

(b) Reasoning about axes as before, show that applying Rotation 3 three times gets us back to where we started.

Solution: We can extend our chart, applying the Rotation 3 chart above over and over.

$$\begin{array}{ccccccc} \text{positive } x\text{-axis} & \mapsto & \text{negative } z\text{-axis} & \mapsto & \text{positive } y\text{-axis} & \mapsto & \text{positive } x\text{-axis} \\ \text{positive } y\text{-axis} & \mapsto & \text{positive } x\text{-axis} & \mapsto & \text{negative } z\text{-axis} & \mapsto & \text{positive } y\text{-axis} \\ \text{positive } z\text{-axis} & \mapsto & \text{negative } y\text{-axis} & \mapsto & \text{negative } x\text{-axis} & \mapsto & \text{positive } z\text{-axis} \end{array}$$

Thus, after three applications of Rotation 3, all axes are back where they started.

(c) If you apply a nontrivial rotation three times and get back to where you started, what is the angle of the rotation?

If the word nontrivial is throwing you off: The issue is that the $0°$ rotation trivially satisfies this condition, since rotating by $0°$ three times yields a $0°$ rotation, which is where we started. But we don't want this answer, since we know that Rotation 3 is not a $0°$ rotation.

Solution: If applying a rotation three times gets us back to where we started, then triple the angle must be a multiple of $360°$. Thus, the angle of the rotation must be a multiple of $120°$.

We can just answer $120°$. Why don't the other multiples of $120°$ matter? Well, $240°$ is equivalent to $-120°$, which is equivalent to a $120°$ rotation about the opposite axis. Meanwhile, $360°$ is equivalent to $0°$, a trivial rotation. Then, $480°$ is equivalent to $120°$, which is our answer. And so forth.

(d) Now we need to find the axis. One way to recognize that a vector is on the axis of rotation is that the rotation will send that vector to itself. Apply this reasoning to find the axis of Rotation 3, using the following hint.

Hint: We can reinterpret the chart for Rotation 1 as follows

$$\text{Rotation 1:}\quad \begin{array}{ccc} \langle a,0,0\rangle & \mapsto & \langle 0,0,-a\rangle \\ \langle 0,b,0\rangle & \mapsto & \langle 0,b,0\rangle \\ \langle 0,0,c\rangle & \mapsto & \langle c,0,0\rangle \end{array}$$

Since rotation is linear, we can add the three lines to find that

$$\text{Rotation 1:}\quad \langle a,b,c\rangle \quad \mapsto \quad \langle c,b,-a\rangle$$

Apply this reasoning to Rotation 3. Then assume that $\langle a,b,c\rangle$ is on the axis of Rotation 3, so $\langle a,b,c\rangle$ gets sent to itself. Solve the resulting equations.

Keep in mind that the axis of rotation is an entire line, so there will be many solutions. For the purposes of this question, you can give your answer by picking one nonzero vector on this line.

Solution: Applying the above reasoning the the chart we found above for Rotation 3, we obtain

$$\begin{array}{rccc} & \langle a,0,0\rangle & \mapsto & \langle 0,0,-a\rangle \\ \text{Rotation 3:} & \langle 0,b,0\rangle & \mapsto & \langle b,0,0\rangle \\ & \langle 0,0,c\rangle & \mapsto & \langle 0,-c,0\rangle \end{array}$$

Adding, we find

$$\text{Rotation 3:} \quad \langle a,b,c\rangle \quad \mapsto \quad \langle b,-c,-a\rangle$$

Assuming that $\langle a,b,c\rangle$ is on the axis, we find that $\langle a,b,c\rangle = \langle b,-c,-a\rangle$. Thus, $a = b$, $b = -c$, and $c = -a$. Concisely, $a = b = -c$. Thus, one point on the axis is $\langle 1,1,-1\rangle$, and any multiple is a valid answer. This matches what we had before, namely $\mathbf{u} = \frac{1}{\sqrt{3}}(\mathbf{i}+\mathbf{j}-\mathbf{k})$.

Technically, based on the work we did here, we haven't figured out whether the rotation is clockwise or counterclockwise. You don't need to worry about this issue for the purposes of this question.